

Differential Equations

Building the Ideas from Scratch

A ground-up companion to ENGR 213, Lecture 1 — written for someone meeting this subject for the first time.

Every example below is worked out step by step, including a few not on the original slides.

0. The One Idea Everything Builds On

You already know what a derivative is: it's a **rate of change**. If y depends on x , then $\frac{dy}{dx}$ just answers “if x nudges up a tiny bit, how much does y nudge?” That's the whole idea — a derivative is a speed of change, whatever the two quantities happen to be. Everything in this subject is built on that one thought.

1. The Big Mental Shift: You're Solving for a Function, Not a Number

In every algebra course, “solve for x ” meant: find a *number*. $2x + 1 = 7 \Rightarrow x = 3$. One number, done.

A differential equation asks a very different kind of question. It doesn't hand you an equation to solve for a number — it gives you a clue about how something *changes*, and asks you to reconstruct the whole story: the function itself.

LET'S DERIVE ONE FROM SCRATCH

Imagine a fast-growing plant, and you're told: “its height grows at a rate equal to twice its current height, in cm per week.” In symbols:

$$\frac{dh}{dt} = 2h$$

This isn't a number for the height — it's a *rule* the height obeys at every instant. The job is to find the function $h(t)$, the entire growth curve, that obeys this rule.

It turns out $h(t) = h_0 e^{2t}$ (where h_0 is the starting height) does the job. Check it:

$$\frac{d}{dt} (h_0 e^{2t}) = 2h_0 e^{2t} = 2h(t) \quad \checkmark$$

You're given a relationship between a function and its own rate(s) of change — and you have to find the function.

2. Where These Equations Actually Come From

Differential equations aren't invented out of thin air — they come from describing, in plain English, how something behaves, and then translating that sentence into symbols. Let's derive two examples from your lecture ourselves.

Coffee cooling down

Think about a hot cup of coffee sitting in a room. From experience, you already know:

- When the coffee is *much* hotter than the room, it loses heat fast.
- As it gets closer to room temperature, it cools more slowly.
- Eventually it stops changing, once it matches the room.

In one sentence: **the rate of cooling is proportional to how far the coffee’s temperature is from the room’s**. Translate that directly into symbols, using T for the coffee’s temperature, T_∞ for room temperature, and a positive constant k for how fast heat escapes:

$$\frac{dT}{dt} = -k(T - T_\infty)$$

The minus sign says: since $T > T_\infty$ (the coffee is hotter), the temperature is *decreasing*. This is Newton’s Law of Cooling, built entirely from common sense — not memorized.

Population growth with limited resources

Start simple: with unlimited food and space, a population grows in proportion to its own size — more organisms means more births. Same structure as the plant: $\frac{dP}{dt} = rP$.

But real environments have a **carrying capacity** K — a maximum sustainable population. As P approaches K , growth should slow down and eventually stop. We need a “braking” factor that:

- is close to 1 (no braking) when P is small compared to K
- is close to 0 (full stop) when P is close to K

The simplest expression that does this is $\left(1 - \frac{P}{K}\right)$. Multiply it onto the simple growth rate:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right)$$

— the logistic growth equation from your slides, arrived at from two sentences of reasoning rather than pulled out of a textbook.

3. Two Words You’ll See Constantly

Every DE relates two kinds of quantities:

- The **independent variable** — the thing you’re free to pick, usually time t or position x .
- The **dependent variable** — the thing being described, which changes *because of* the independent variable. In the plant example, h (height) depends on t (time).

Rule of thumb: whatever’s being differentiated (as in $\frac{dh}{dt}$) is the dependent variable; whatever’s on the bottom of that fraction is the independent one.

4. Why We Classify Equations At All

Before solving anything, it helps to know *what kind* of equation you’re facing — the same way “solve $2x = 6$ ” needs a different approach than “solve $x^2 = 6$.” Differential equations get sorted

along three axes — **Type, Order, Linearity** — and each one tells you which solving techniques will actually work.

4.1 Type — how many “directions” of change are involved?

- Only one independent variable (like t alone) → **ordinary** differential equation (ODE). This whole course is about these.
- Two or more (like both x and t) → **partial** differential equation (PDE) — these show up when something depends on both time *and* position, like heat spreading through a metal rod. PDEs get their own later course; you just need to recognize the difference for now.

4.2 Order — how many derivatives deep does it go?

Think about a car:

- position $x(t)$
- velocity $= \frac{dx}{dt}$ (also written x') — one derivative
- acceleration $= \frac{d^2x}{dt^2}$ (also written x'') — two derivatives

If an equation only involves velocity, it's **1st order**. If it involves acceleration, it's **2nd order** — order only looks at the *highest* derivative present, never at how many derivatives appear in total, and never at what power any of them is raised to.

EXAMPLE

$$\frac{d^2y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 - 4y = e^x \rightarrow \text{2nd order}$$

$\left(\frac{dy}{dx} \right)^3$ is cubed, but $\frac{dy}{dx}$ itself is only a first derivative. The highest derivative anywhere in the equation is $\frac{d^2y}{dx^2}$ — so the whole equation is 2nd order.

4.3 Linearity — is y tangled up with itself?

A DE is **linear** if every term is just (some expression in x) times y or one of its derivatives, all added together:

$$a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

Quick test — ask: *is y ever squared, stuck inside a sin or $e^{(\cdot)}$, or multiplied by its own derivative?* If yes, it's **nonlinear**.

$$\underbrace{\frac{dy}{dx} = 3y}_{\text{linear — just a constant times } y}$$

$$\underbrace{\frac{dy}{dx} = y^2}_{\text{nonlinear — } y \text{ is squared}}$$

WHY BOTHER? Linear equations have well-developed, systematic solving methods — most of this course teaches you those. Nonlinear equations are often much harder; sometimes there's

no exact formula at all, and you have to approximate the answer numerically instead.

5. What Does “Solving” (and “Checking”) Actually Mean?

Just like you can check whether $x = 3$ solves $2x + 1 = 7$ by plugging it in, you check whether a *function* solves a differential equation by substituting the function — and its derivatives — back in, and seeing if the equation becomes true.

THE SIMPLEST POSSIBLE CASE

$$\frac{dy}{dx} = 2x$$

This just asks: what function has slope $2x$ everywhere? From basic integration, $y = x^2$ has derivative $2x$. Check: $\frac{d}{dx}(x^2) = 2x$ ✓. (We’ll see shortly that $y = x^2 + C$ works too, for *any* constant C .)

A CASE FROM YOUR LECTURE

$y'' + y = 0$, try $y = \sin(x)$.

Derivatives: $y' = \cos(x)$, $y'' = -\sin(x)$. Substitute back: $y'' + y = -\sin(x) + \sin(x) = 0$ ✓ — valid for every real x .

A CAUTIONARY CASE — CHECKING ISN’T JUST ALGEBRA

$y'' = \frac{2}{x^3}$, try $y = \frac{1}{x}$.

Derivatives: $y' = -\frac{1}{x^2}$, $y'' = \frac{2}{x^3}$ — matches perfectly ✓ — *except* at $x = 0$, where $y = \frac{1}{x}$ isn’t even defined. So this solution only holds on $(-\infty, 0)$ or $(0, \infty)$, never across $x = 0$.

LESSON: a solution isn’t just a formula that satisfies the equation — it also comes with an *interval* on which it actually makes sense.

6. Explicit vs. Implicit — Why We Sometimes Don’t Isolate y

Sometimes you can write “ $y =$ (a formula in x)” — that’s **explicit**. Other times, the natural answer is a relationship between x and y that you deliberately leave tangled — that’s **implicit**. Here’s why, derived from a shape you already know.

DERIVING WHY A CIRCLE STAYS IMPLICIT

Take $x^2 + y^2 = 25$. Solve for y :

$$y = \pm\sqrt{25 - x^2}$$

That $\pm$ means this isn’t *one* function — it’s two (the top half of the circle, and the bottom half). A proper function can only give one output per input, so we usually just leave it as

$x^2 + y^2 = 25$ rather than force it into two explicit pieces.

You can still check it against a DE using **implicit differentiation**: $\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25) \Rightarrow 2x + 2y y' = 0 \Rightarrow y' = -\frac{x}{y}$. So $x^2 + y^2 = 25$ is an implicit solution of $\frac{dy}{dx} = -\frac{x}{y}$.

7. Families of Solutions — Why Is There Always a Mystery Constant?

STARTING FROM THE SIMPLEST DE THERE IS

$$\frac{dy}{dx} = 0$$

What functions have a derivative of zero everywhere? Any **constant** function: $y = 5$, $y = -2$, $y = 100$ — in fact $y = C$ for *any* number C . This is a **one-parameter family of solutions**: a whole family of answers, indexed by one constant.

Bump it up slightly — $\frac{dy}{dx} = 2x$ — and the same idea gives $y = x^2 + C$: one arbitrary constant, because solving a first-order equation is essentially one integration, and every integration introduces one constant.

A **second-order** equation needs **two** integrations to unwind, so it naturally comes with **two** arbitrary constants:

$$y'' - 2y' + y = 0 \quad \longrightarrow \quad y = c_1 e^x + c_2 x e^x$$

An n th-order differential equation typically has an n -parameter family of solutions — one constant for every “layer” of differentiation you have to undo.

8. Systems — When Two Things Change Together

Sometimes one quantity’s rate of change depends on *another* quantity, and vice versa, so neither can be solved alone. The classic intuitive picture: **rabbits and foxes** sharing an ecosystem.

- The rabbit population grows on its own, but shrinks the more foxes are around (foxes eat rabbits).
- The fox population grows based on how many rabbits are available to eat, but shrinks on its own over time.

Neither equation stands alone — you need both, together, as a **system**:

$$\frac{dx}{dt} = f(t, x, y) \quad \frac{dy}{dt} = g(t, x, y)$$

A solution to a system is a *pair* of functions satisfying both equations at once, on the same interval. Here’s a concrete (linear) example from your lecture:

EXAMPLE

System: $\frac{dx}{dt} = x + 3y$, $\frac{dy}{dt} = 5x + 3y$

Solution: $x = e^{-2t} + 3e^{6t}$, $y = -e^{-2t} + 5e^{6t}$

Checked the same way as always — differentiate each proposed function and confirm it matches the right-hand side of its own equation.

The Big Picture, in One Paragraph

A differential equation is a sentence about how something changes, translated into math. Solving it means finding the actual function that behaves that way — not a number, a whole curve. Before solving, we classify the equation (type / order / linearity) because that tells us which tools apply. Once we have a candidate solution, we check it by substituting it — and its derivatives — back in, and we note the actual interval where it's valid. And because differentiating erases constants, solutions almost always come in families, with one arbitrary constant for every order of the equation.

Companion to: ENGR 213, Lecture 1 (Sept 9, 2026), Prof. A. Haghighat M. — reference text: Zill, *Advanced Engineering Mathematics* (7th ed.), Section 1.1. Examples marked “let’s derive” are original, built to illustrate the same ideas from the lecture.